
EC2.103: Analog Electronic Circuits - Spring 2026

Assignment - 3: MOS Basics & Amplifiers

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Instructions:

1. The assignment must be submitted as a single PDF file named **Name_RollNo.pdf** through Moodle.
 2. The report must be self-explanatory and include complete solutions. Each answer should contain the required schematics, SPICE directives, annotated simulation waveforms, and a clear discussion or inference based on the obtained results.
 3. All circuit simulations must use the **nmos4 & pmos4** in **TSMC 180nm** tech node for MOS device models wherever required.
 4. Your **Roll Number must be clearly displayed on every schematic and simulation plot**. Failure to do so will result in a **zero for the respective question**.
 5. All queries related to the assignment must be posted on Moodle. Discussions are encouraged; however, they must take place strictly on the Moodle forum.
 6. **Any form of plagiarism, copying, or academic misconduct will result in an immediate F grade for the course.**
 7. The assignment must be submitted on or before **04/04/2026** at **11:59 PM**. **No extensions will be granted under any circumstances.**
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Q1. Consider an NMOS transistor with $\frac{W}{L} = \frac{1.8\mu m}{0.18\mu m}$. Use the TSMC 180nm technology file for all simulations.

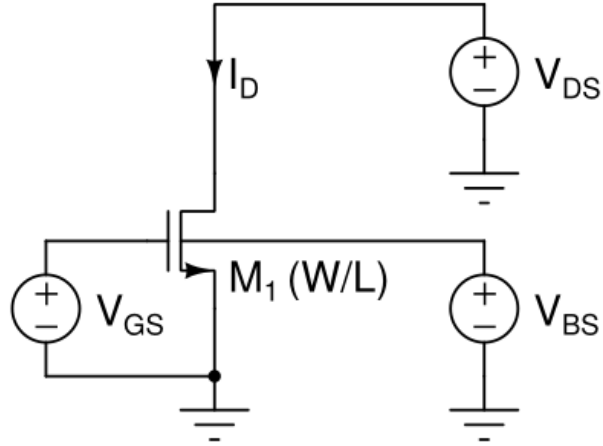
a. I_D vs V_{DS} Characteristics

As shown in Fig. **a.**, plot I_D vs V_{DS} .

Sweep:

- V_{DS} from 0 to 1.8 V in steps of 0.01 V
- V_{GS} from 0 to 1.8 V in steps of 0.3 V

Assume $V_{BS} = 0$ V.



Hint: Use NMOS4 with model name CMOSN. Define:

$$\begin{aligned}
 AS &= 5 \cdot W \cdot \lambda \\
 AD &= 5 \cdot W \cdot \lambda \\
 PS &= 10\lambda + 2W \\
 PD &= 10\lambda + 2W
 \end{aligned}$$

where $2\lambda = L$.

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.include TSMC_180nm.txt
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b. Extraction of μC_{ox} and V_T

Plot I_D vs V_{GS} for:

- $V_{DS} = 50 \text{ mV}$
 - $V_{BS} = 0 \text{ V}$
- i. Estimate μC_{ox} and V_T from the graph.
 - ii. Plot I_D vs V_{GS} for $V_{DS} = 1.8 \text{ V}$ and extract V_T .
 - iii. Compare the two V_T values and explain any difference (Hint: DIBL).

c. Effect of Body Bias on Threshold Voltage

Using I_D vs V_{GS} simulations, determine V_T for NMOS and PMOS devices with:

$$\frac{W}{L} = \frac{1.8\mu m}{0.18\mu m}$$

Perform simulations for:

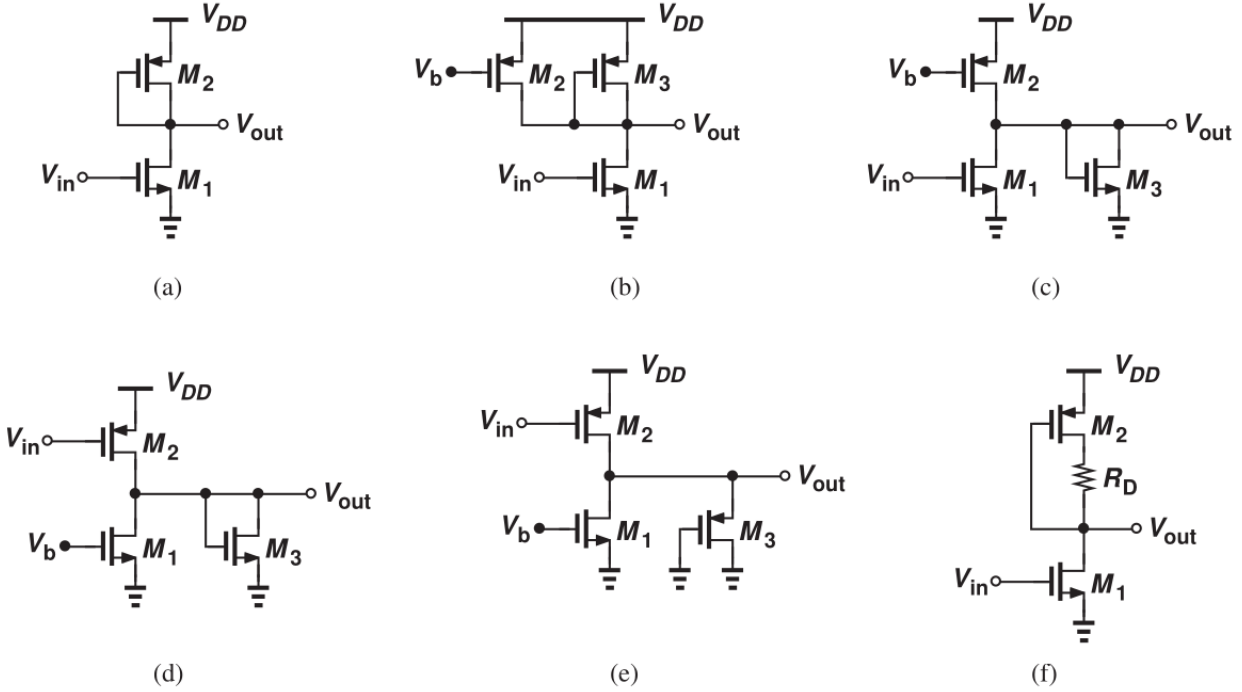
- $V_{BS} = 0 \text{ V}$
- $V_{BS} = +900 \text{ mV}$
- $V_{BS} = -900 \text{ mV}$

Overlay the three plots and:

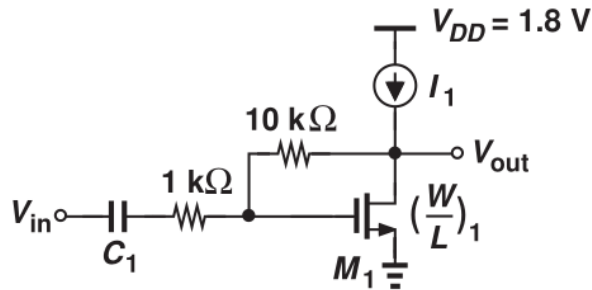
- Compare the extracted V_T values
- Briefly discuss the observed differences (Hint: Body effect)

Note: For PMOS simulations, use PMOS4 with model name CMOSP.

Q2. Consider the presence of Early Effect and determine the voltage gain of the stages shown in Fig. Q2..

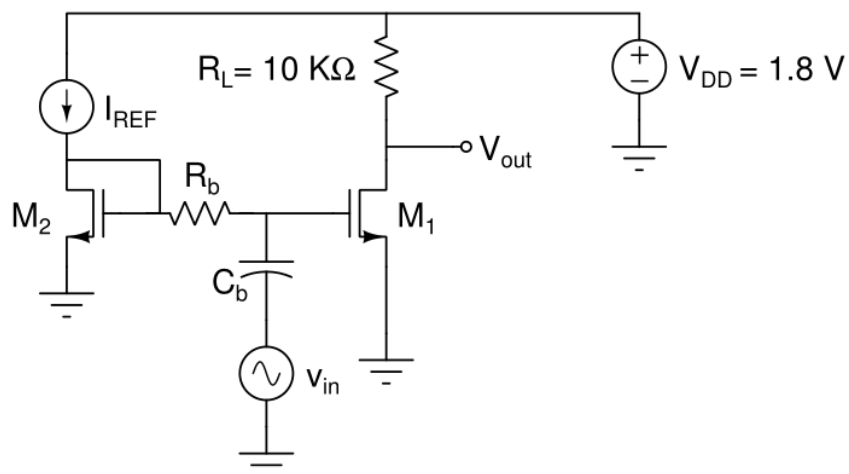


Q3. In the circuit of Fig.Q3., I_1 is an ideal current source equal to 1 mA.



- Using hand calculations, determine $(W/L)_1$ such that $g_{m1} = (100 \Omega)^{-1}$.
- Select C_1 for an impedance of $\approx 100 \Omega (\ll 1 \text{ k}\Omega)$ at 50 MHz.
- Simulate the circuit and obtain the voltage gain and output impedance at 50 MHz.
- What is the change in the gain if I_1 varies by $\pm 20\%$?

- Q4.** Design a common source amplifier (shown in Fig. **Q4.**) with a resistive load of $10\text{ k}\Omega$ for a voltage gain > 5 and an overdrive voltage (of input transistor) of 200 mV . The minimum input signal frequency is 100 Hz . Design for minimum power consumption. Clearly state your assumptions (if any).



- Show the design procedure with calculations for sizes of transistors, I_{REF} , C_b , and R_b . What is the overall power consumed by your amplifier?
- Provide transient simulation plots (4–5 cycles) showing the gain, considering

$$v_{in} = 10 \sin(2\pi \cdot 1000 t) \text{ mV}.$$

- Show the AC response plots ($20 \log |A_v|$ vs frequency) and determine the unity gain bandwidth (f_u). Sweep frequency from 1 Hz to 1 GHz .